

Space-Based AI Computing: Physically Feasible, Engineering-Fragile, Commercially Distant

Three-Tier Independent Verification Based on Space Solar and
NVIDIA Architecture

Characterizing the Disagreement: It's Not About Whether, It's About How Fast

Jensen Huang (conservative & prudent) vs. Elon Musk (extremely optimistic)—the disagreement is not about "who miscalculated"

- Time window, power sizing, system boundary—three variables differ, conclusions naturally diverge
- Core judgment: it's not about whether, but **how fast**

Three-Tier Analysis Overview

Physically Feasible—S-B Law and radiative heat dissipation self-consistent

Engineering-Fragile—mass chain and manufacturing margins tight

Commercially Distant—TCO gap vs. ground: **11.8×**

Heat Dissipation Feasible: S-B Law and 1,400 W/m² Operating Point

Stefan-Boltzmann law:

$$\frac{P}{A} = 2\varepsilon\sigma T^4$$

$$\sigma = 5.6704 \times 10^{-8} \text{ W} \cdot \text{m}^{-2} \cdot \text{K}^{-4}$$

Al1: Using $\varepsilon \approx 0.90$, $T \approx 342.5 \text{ K}$ (69.5°C), yields **$\sim 1,400 \text{ W/m}^2$**

Physically self-consistent, but approaching coating EOL degradation margin

LEO Is Not a Cold Sink: Eclipse and Thermal Environment Constraints

Three heat sources combined: direct solar **1,361 W/m²** + Earth albedo **~122 W/m²** + Earth IR **~240 W/m²**

- Net heat dissipation capacity **15–40%** lower than ideal vacuum
- Eclipse **35.5 min** / orbit **96.7 min**, requires **71.0 kWh** battery

Conclusion: LEO is not an infinite cold sink

Satellite Cost Breakdown: Where the \$83.61M Goes

9 subsystems, satellite **8.48 t**

- Photovoltaic array **\$46.27M (55.3%)**—largest single item
- System margin/redundancy **\$18.29M (21.9%)**
- Per-satellite manufacturing cost **\$83.61M**
- Launch cost only **~1.8%** of TCO

Three-Number Contradiction: 120 / 150 / ~210 kW

- **120 kW**—claimed sustained IT load
- **150 kW**—GPU peak power
- **~210 kW**—array EOL physical requirement derived from LEO day-night cycle:
eclipse-segment battery charging + sunlit-segment IT +
heat dissipation continuous power—closes at **~210 kW**

Conclusion: Both the 120 kW claimed value and 150 kW peak power exhibit a **1.4× deviation** from the **~210 kW** physical requirement of orbital energy balance—a physics-level contradiction

Launch Scale Reality Check: 300 GW/yr vs. FAA Cap of 69 Launches/yr

Musk claims **300–500 GW/yr** deployment capacity

- FAA-approved Starship annual launch cap: **69 launches**
- At 10 satellites/launch, 120 kW/satellite: **~0.083 GW/yr**
- Gap: **~3,600× (3 orders of magnitude)**

Even with substantial FAA cap relaxation,
combined improvement of all three parameters
cannot close the order-of-magnitude gap

Lifetime Leverage: 5yr HJT×2 Outweighs 10yr GaAs

- 10yr-GaAs (single generation): **\$764.98M**
- 5yr-HJT×2 (requires re-launch): **~\$1,005M (+31.4%)**
- HJT cells are 70–100× cheaper, but **2× re-launch** cost outweighs
- Orbital GPU AFR **5–15%/yr**, maintenance infeasible

Conclusion: On-orbit lifetime is a more important economic lever than cell cost

Three-Branch TCO Comparison: Ground vs. Space

Scenario	10yr Total Cost	Gap vs. Ground
Ground IDC	\$64.6M	1 × (baseline)
10yr-GaAs (primary)	\$764.98M	11.8×
5yr-HJT×2 (gap-filling estimate)	~\$1,005M	15.6×
5yr-GaAs×2 (control)	\$1,341M	20.8×

Manufacturing (especially photovoltaic array) is the core TCO driver

Commercially Distant: \$765M vs. \$65M
= 11.8×

10yr-GaAs **\$764.98M** vs. ground **\$64.6M**

Gap = **11.8×**

Cost Anatomy: Launch Is Only 1.8%

- Manufacturing cost **91.1%**—absolutely dominant
- Photovoltaic array **50.4%**, system margin **20.0%**
- Launch only **1.8%**—less than \$2 out of every \$100 goes to launch

Directional leverage is in manufacturing, not launch

Reversal Conditions: Four-Parameter Joint Probability < 25%

Four parameters met simultaneously:

lifetime $\geq 7\text{yr}$ + PV < \$50/W_{BOL} + AIT Starlink-class + launch < \$100/kg

- Independent joint probability ~**8.4%**, after positive-correlation adjustment <**25%**

Conclusion: possible but uncertain,
insufficient to support a definitive commercial commitment

Timeline: 2026 → 2028 → 2030 → 2035

Phase	Time	Assessment
Prototype Verification	2026–2028	Commercially infeasible, technology validation meaningful
Niche Deployment	2028–2030	Military / low-latency inference scenarios
Conditional Expansion	2030–2035	Four-condition joint probability <25%

Summary

Known Facts · Computable Chains · Currently Undecidable

Physical foundation is solid; commercial commitment should not be made prematurely